TENGHAN ZHONG

tenghanz@usc.edu | +1 213-507-2381

EDUCATION

University of Southern California (USC) Sep. 2024 – May 2026
Master of Science in Mathematical Finance Los Angeles, CA
USC Dornsife College
GPA: 4.0/4.0

Nanchang University – Project 211 Sep. 2020 – Jun. 2024
Bachelor of Economics in Financial Mathematics Nanchang, China
School of Mathematics and Computer Science
GPA: 3.62/4.0; 87.02/100 · **Class rank:** 2/25

Graduate-level coursework: Stochastic Calculus and Mathematical Finance A & B; Mathematical Statistics (PhD level); Linear Programming (PhD level); Machine Learning in Quantitative Finance; Financial Informatics and Simulation; Time Series Analysis; Big Data Econometrics; Investment Analysis and Portfolio Management.

PUBLICATIONS

Tenghan Zhong. Quantifying Repair Risk in Crypto Option Surfaces. *ACM International Conference on AI in Finance (ICAIF), 2026*. Selected for oral presentation.

- Identified and quantified how option surfaces fitted to the same quotes under bid–ask bounds and no-arbitrage constraints can produce different Greeks and hedge positions; derived a locally sharp bound on curvature amplification and used linear programming to determine the feasible curvature range.
- Demonstrated on Deribit BTC and ETH options that surface-construction choices materially change hedge quantities and risk estimates even when all resulting surfaces satisfy the same constraints.

Tenghan Zhong, Yuxuan Ouyang, and Keyuan Wu. FNUG: Factor-Neutral Update Gating for Anchored Option Hedging. *ACM International Conference on AI in Finance (ICAIF), 2026*. Accepted for poster presentation (22% acceptance rate).

- Developed FNUG, a hedging framework that uses option-chain information to learn adjustments around a least-squares delta–vega hedge while approximately preserving its exposures.
- Achieved the lowest 95% CVaR among all evaluated classical and learning-based hedging methods in the primary SPX setting with transaction costs.

Tenghan Zhong and Keyuan Wu. Marking-Aware Sequential VaR Recalibration for Standardized Option Books. *IEEE Symposium on Computational Intelligence for Financial Engineering and Economics (CIFEr), 2026*, 90–97. Selected for long presentation. [DOI]

- Developed a VaR framework that incorporates marking rules into the definition of option-book losses and designed a sequential recalibration method to correct undercoverage using past forecast residuals.
- Brought exceedance rates close to the target across all six SPX and QQQ book–market pairs, with the lowest average loss beyond predicted VaR among evaluated methods in both markets.

Tenghan Zhong. Risk-Sensitive Specialist Routing for Volatility Forecasting. *IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2026*. Regular paper. [arXiv]

- Developed a risk-sensitive forecasting method that selects and combines volatility models based on relative predictive performance, market-state similarity, and penalties for underprediction.
- Achieved the lowest overall median forecast loss on six ETFs among evaluated methods; reduced median forecast and underprediction losses by 24% and 22% in high-volatility periods relative to selecting the best recent performer.

Tenghan Zhong. Feature or Penalty? A Placement Diagnostic for Graph-Risk Signals in Learned Portfolio Decisions. *ACM International Conference on Information and Knowledge Management (CIKM), 2026*. Accepted (short paper).

- Designed controlled experiments to distinguish the value of graph-risk information from the effect of incorporating it into portfolio contracts as an exposure penalty.
- Showed on industry portfolios and ETFs that the penalty itself explained aggregate CVaR reductions, while graph-based asset ordering contributed more clearly to reducing industry crowding.

Tenghan Zhong, Keyuan Wu, and Yuxuan Ouyang. Public-Disclosure Recoverability in Fund Allocation Recommendations. *ACM International Conference on AI in Finance (ICAIF), 2026*. Selected for oral presentation.

- Inferred empirical allocation bounds from 295,182 regulatory fund reports and designed an evaluation to distinguish the predictive value of learned rankings from portfolio persistence and limitations in candidate coverage.
- Found that a benchmark favoring prior holdings outperformed all evaluated learned rankers within those bounds, while large reallocations often lacked nearby feasible candidates, limiting what ranking alone could recover.

RESEARCH EXPERIENCE

Research Assistant – Quantitative Research Group Jun. 2025 – Sep. 2025
University of California, Los Angeles Los Angeles, CA
Supervisor: Prof. Guang Cheng

- Constructed a synchronized cross-market dataset of 5M+ minute-level observations linking Polymarket prediction contracts with Deribit options for pricing and microstructure analysis.
- Developed option-replication benchmarks for Polymarket interval contracts using matched Deribit vertical spreads; found persistent cross-market pricing discrepancies after accounting for transaction costs and settlement timing.
- Developed an in-play MLB decision model using game-state and market-microstructure signals; decomposed performance across market conditions to identify where predictive edge persisted after execution costs.

Undergraduate Thesis: Dynamic Pricing of ETF Options Based on Recurrent Neural Networks
Nanchang University; Supervisor: Prof. Lidan Liao Dec. 2023 – Feb. 2024

- Proposed a structural-residual pricing framework that integrates Heston pricing with GRU/LSTM residual correction, reducing out-of-sample pricing MSE and MAPE by at least 70% relative to structural-model benchmarks.
- Identified persistent temporal dependence in Heston pricing residuals through statistical dependence tests and rolling out-of-sample analysis, providing empirical support for the residual-learning design.

PROJECT EXPERIENCE

Calibration of Power-Type Derivatives for Rough Volatility with Jumps
USC; Supervisor: Prof. Weixuan Xia [Code] Feb. 2025 – May 2025

- Developed a neural-network-assisted calibration method for power-type volatility derivatives, reducing calibration time from over 24 hours to under 20 minutes and MAE from 0.047 to 0.033.
- Used high-frequency activity-index estimates to identify the pure-jump regime and inform parameter restrictions for more stable calibration across maturities.

Hybrid Approaches to Crypto Volatility Prediction Using GARCH and GRU Models
USC; ECON 515 course project; Instructor: Prof. Ratika Narag [Code] Sep. 2024 – Dec. 2024

- Developed two hybrid GRU–GARCH architectures for crypto-volatility forecasting; evaluated them on 830,000+ five-minute Coinbase observations, with the AVARCH-based hybrid achieving lower forecast MAPE than GARCH-family benchmarks and standalone GRU across T+1 to T+6 horizons.
- Derived likelihood gradients and Hessians for GARCH, GJR-GARCH, EGARCH, and AVARCH to analyze local curvature and investigate estimation difficulties observed in cryptocurrency data.

TEACHING EXPERIENCE

University of Southern California – Grader Sep. 2025 – May 2026
Dornsife College of Letters, Arts and Sciences Los Angeles, CA
Department of Mathematics: MATH 125, Calculus I (Fall 2025); MATH 125, Calculus I (Spring 2026).

PROFESSIONAL EXPERIENCE

Vanho Securities Co., Ltd. Sep. 2023 – Nov. 2023
Quantitative Research Intern, Asset Management Department Shenzhen, China

- Reproduced and extended timing and value-stock-selection strategies from Everbright and Huatai Securities research, building a unified Python backtesting framework with performance attribution and risk analysis.
- Refined strategy parameters and portfolio constraints through sensitivity analysis. In out-of-sample backtests, the revised strategies increased excess returns by approximately 200% and reduced maximum drawdown by 20% relative to the reproduced baseline.

Ping An Securities Co., Ltd. Jul. 2022 – Aug. 2022
Investment Product Analyst Intern, Brokerage Division Shenzhen, China

- Processed and validated private-equity income and product-level financial data using Excel and Python, supporting periodic performance monitoring and reporting.
- Prepared weekly and monthly reports on private-equity products, analyzing operating performance, return trends, and key portfolio metrics.

HONORS AND AWARDS

Nanchang University
First-Class Scholarship Sep. 2023
Outstanding Student Leader Mar. 2023
First-Class Scholarship Sep. 2022

SKILLS AND LANGUAGES

Programming: Python, R, SQL
Tools: Git/GitHub, PostgreSQL, LaTeX, Microsoft Office
Languages: Chinese (native), English (proficient)